Practical 7

Aim: Implement a Three-Address Code Generator for Control Statements: Write semantic rules to generate three-address code for control statements. (Use C/C++ language.)

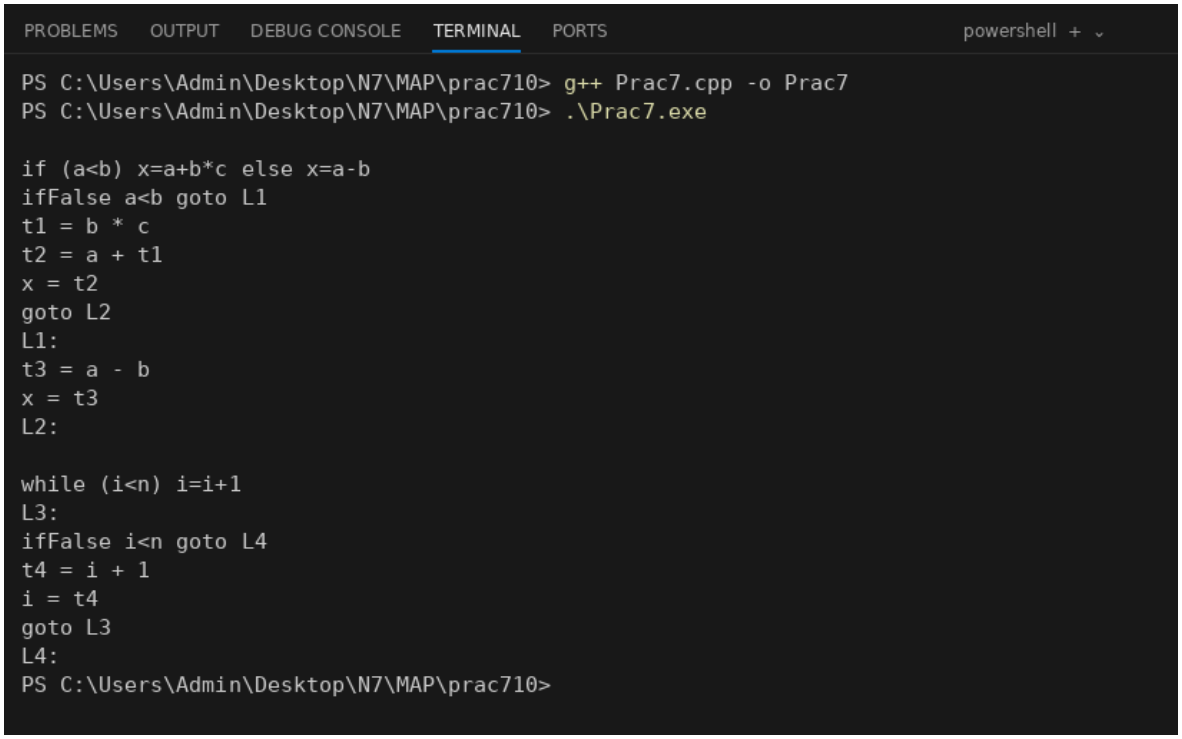
Prac7.cpp:

```
#include <iostream>
#include <vector>
#include <string>
using namespace std;

int t = 0, l = 0;

void gen(string s)
{
    string x = s.substr(0, 1);
    vector<string> e;
    for (int i = 2; i < s.size(); i++)
        e.push_back(string(1, s[i]));

    for (int i = 1; i < e.size(); i += 2)
    {
        if (e[i] == "*" || e[i] == "/")
        {
            string temp = "t" + to_string(++t);
            cout << temp << " = " << e[i-1] << " " << e[i] << " " << e[i+1] << endl;
            e[i-1] = temp;
            e.erase(e.begin() + i, e.begin() + i + 2);
            i -= 2;
        }
    }
    for (int i = 1; i < e.size(); i += 2)
    {
        string temp = "t" + to_string(++t);
        cout << temp << " = " << e[i-1] << " " << e[i] << " " << e[i+1] << endl;
        e[i+1] = temp;
    }
    cout << x << " = " << e.back() << endl;
}

void ifElse(string cond, string s1, string s2)
{
    int L1 = ++l, L2 = ++l;
    cout << "\nif (" << cond << ") " << s1 << " else " << s2 << endl;
    cout << "ifFalse " << cond << " goto L" << L1 << endl;
    gen(s1);
    cout << "goto L" << L2 << endl;
    cout << "L" << L1 << ":" << endl;
    gen(s2);
    cout << "L" << L2 << ":" << endl;
}

void whileLoop(string cond, string s)
{
    int L1 = ++l, L2 = ++l;
    cout << "\nwhile (" << cond << ") " << s << endl;
    cout << "L" << L1 << ":" << endl;
    cout << "ifFalse " << cond << " goto L" << L2 << endl;
    gen(s);
```

```
    cout << "goto L" << L1 << endl;
    cout << "L" << L2 << ":" << endl;
}

int main()
{
    ifElse("a<b", "x=a+b*c", "x=a-b");
    whileLoop("i<n", "i=i+1");
    return 0;
}
```

Output:

An if-else statement and a while loop are converted to TAC using labels and jumps.



For x=a+b*c, b*c is computed first into t1 because * has higher precedence. In the while loop, goto L1 jumps back to recheck the condition.